

HUISEOK MOON

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RESEARCH INTERESTS

Assistive robotics and physical human–robot interaction, grounded in clinically validated real-time intent detection and sensor-based assistive control (EMG, IMU). Current focus: physics-based modeling and simulation, model-based control, and multimodal AI for understanding and predicting human motion in wearable robotics.

Keywords: Wearable Robotics • Assistive Robotics • Multimodal Learning • Nonlinear Control

EDUCATION

University Paris-Est Créteil

Ph.D. in Signal, Image and Control — Advisor: Dr. Samer Mohammed

Thesis: Human-in-the-loop Paradigms for Mobility Assistance Using Powered Exoskeleton

Vitry-sur-Seine, France

Nov 2019 – Jan 2024

Sogang University

M.S. in Mechanical Engineering — Advisor: Dr. Kyoungchul Kong

Thesis: Control Algorithm of a Robotic Leg for Reducing Interaction Force Between a Load and the Human Body

Graduate Research Fellowship, Ilwun Science and Technology Foundation (2017–2019)

Seoul, South Korea

Mar 2017 – Feb 2019

Kwangwoon University

B.S. in Robotics (Intelligence System)

Thesis: Research on Personal Mobility Using Ball Balancing Robot

Seoul, South Korea

Mar 2010 – Feb 2017

PROFESSIONAL EXPERIENCE

Université Paris-Est Créteil

Postdoctoral Researcher

Vitry-sur-Seine, France

Jul 2026 – Present

- Designed a high-precision gait mode recognition system using LSTM networks, achieving 98% classification accuracy and enabling real-time intent detection for actuated ankle-foot orthoses.
- Developed an energy-efficient adaptive FES control strategy based on surface EMG signals, reducing stimulation energy consumption by 30% while mitigating muscle fatigue in clinical trials.
- Implemented a 7-IMU multi-sensor fusion framework extracting kinematic metrics for lower-limb joint analysis and Parkinson's disease diagnosis in a portable clinical device.

Korea Advanced Institute of Science & Technology (KAIST)

Postdoctoral Researcher — Advisor: Dr. Jaegul Choo

Daejeon, South Korea

Sep 2025 – Mar 2026

- Developed physics-informed AI models for human motion intention inference, leveraging motion-primitive representations of wearable IMU signals.

Université Paris-Est Créteil

Research Engineer

Vitry-sur-Seine, France

Sep 2023 – Aug 2025

- Built the LSTM-based gait mode recognition pipeline (98% accuracy) underpinning real-time control of actuated ankle-foot orthoses.
- Devised an adaptive EMG-driven FES control strategy, cutting stimulation energy by 30% and reducing muscle fatigue in clinical validation.

Université Paris-Est Créteil

ATER (Teaching & Research Assistant)

Vitry-sur-Seine, France

Nov 2022 – Aug 2023

- Delivered undergraduate lectures and labs on wearable sensor integration (EMG, IMU, FSR) and biomechanical signal processing in Python and MATLAB.

Université Paris-Est Créteil

Research Assistant

Vitry-sur-Seine, France

Nov 2019 – Oct 2022

- Developed nonlinear control architectures (impedance control, reference tracking) for human–exoskeleton interaction, applied to lower-limb exoskeletons and sit-to-stand assistance robots.
- Constructed a real-time activity recognition pipeline combining Random Forests and deep learning with Kalman-filtered IMU features for intent-based mobility assistance control.

Sogang University

Research Assistant

Seoul, South Korea

Mar 2017 – Feb 2019

- Designed interaction force control strategies for lower-limb exoskeletons to augment strength and endurance in load-carrying tasks.

PUBLICATIONS

Journal Articles

- [J1] X. Luo, **H. Moon**, J.-M. Gracies, W. Huo and S. Mohammed. (2026). Knee-Guided Phase–Magnitude Decoupled Adaptive Functional Electrical Stimulation for Drop-Foot Correction. *IEEE Trans. Neural Syst. Rehabil. Eng.*, (Submitted) Equally contributed as first author and corresponding author
- [J2] L. Marusu, C. Gault-Colas, **H. Moon**, S. Mohammed, J.-M. Gracies (2026) Changes in Walking Speed, Step Length, and Lower Limb Kinematics Over Eight Weeks of Guided Self-Rehabilitation Contracts in Hemiparetic Patients: A Prospective Cohort Study. *J. Neural Eng.*, (Submitted)
- [J3] O. Bey, M. Chemachema, R. Jradi, **H. Moon**, H. Rifai, Y. Amirat. S. Mohammed (2026). Non-model-based Finite-Time Adaptive Neural Output Feedback Control for an Active Ankle-Foot Orthosis. *IEEE Trans. Autom. Sci. Eng.*, 41, 4794–4809. [DOI]
- [J4] **Moon, H.**, Bey, O., Boubezoul, A., Oukhellou, L., Mohammed, S. (2025). Real-time LSTM-driven dynamic gait mode detection for enhanced control of actuated ankle-foot orthosis. *IEEE Trans. on Robotics*, [DOI]
- [J5] Khodabandelou, G., **Moon, H.**, Amirat, Y., Mohammed, S. (2023). Fuzzy convolutional attention-based GRU network for human activity recognition. *Eng. Applications of Artificial Intelligence*, 118, 105702. [DOI]
- [J6] **Moon, H.**, Maiti, R., Das Sharma, K., Amirat, Y., Siarry, P., Mohammed, S. (2022). Hybrid half-Gaussian selectively adaptive fuzzy control of an actuated ankle-foot orthosis. *IEEE Robotics and Automation Letters*, 7, 9635–9642. Presented at ICRA 2023, oral. [DOI]
- [J7] Jammeli, I., Chemori, A., **Moon, H.**, Elloumi, S., Mohammed, S. (2022). An assistive explicit model predictive control framework for a knee rehabilitation exoskeleton. *IEEE/ASME Trans. on Mechatronics*, 27, 3636–3647. [DOI]
- [J8] Huo, W., **Moon, H.**, Alouane, M. A., Bonnet, V., Huang, J., Amirat, Y., Vaidyanathan, R., Mohammed, S. (2022). Impedance modulation control of a lower-limb exoskeleton to assist sit-to-stand movements. *IEEE Trans. on Robotics*, 38, 1230–1249. [DOI]

Conference Proceedings

- [C1] Bey, O., Jradi, R., **Moon, H.**, Rifai, H., Das Sharma, K., Amirat, Y., Mohammed, S. (2024). A novel funnel-based L1 adaptive fuzzy approach for the control of an actuated ankle foot orthosis. *IEEE ICRA*. [DOI]
- [C2] **Moon, H.**, Boubezoul, A., Oukhellou, L., Amirat, Y., Mohammed, S. (2022). Online human intention detection through machine-learning based algorithm for the control of lower-limbs wearable robot. *IEEE-RAS Humanoids*. [DOI]
- [C3] Jung, P.-G., Huo, W., **Moon, H.**, Amirat, Y., Mohammed, S. (2021). A novel gait phase detection algorithm for foot drop correction through optimal hybrid FES-orthosis assistance. *IEEE ICRA*. [DOI]

ACADEMIC SERVICE AND TEACHING

Peer Review	IEEE Trans. on Robotics, IROS, ICRA, Eng. Applications of AI, Mechatronics, and others
Invited Talks	REHAssist Visiting Seminar, EPFL, Switzerland (2025); LISSI–BIOTN Joint Seminar, UPEC, France (2024); LISSI Research Seminars, UPEC, France, 4 sessions (2020–2023)
Volunteering	IFAC World Congress, volunteer (2025); IUT Research Day, poster exhibition (2024); Fête de la Science, public outreach exhibitor (2023); Korea Defense Science & Tech Fair, exhibitor (2017)
Mentoring	M.S. students: L. Marusu (2025), A. A. Benhamlaoui (2024), D. Plet (2024), W. Legendre (2023), J. M. Fadous (2022), R. Ravaka & M. Diaby (2021) — topics spanning multi-sensor gait analysis, AI-based gait event detection, Parkinson’s diagnostics, and assistive robot control

SKILLS

Programming	Python, C, C++, MATLAB, LabVIEW, SolidWorks
Frameworks	PyTorch, TensorFlow, Simulink
Expertise	Real-time ML deployment, multi-modal sensor fusion, embedded systems, adaptive & impedance control, human–robot interaction, wearable robotics, signal processing
Languages	Korean (native), English (fluent), French (basic)

REFERENCES

Dr. Jaegul Choo Associate Professor Kim Jaechul Graduate School of AI, KAIST <i>Postdoc Advisor</i> jchoo@kaist.ac.kr	Dr. Samer Mohammed Associate Professor LISSI Laboratory, Université Paris-Est Créteil <i>Ph.D. Advisor</i> samer.mohammed@u-pec.fr	Dr. Kyoungchul Kong Associate Professor KAIST <i>M.S. Advisor</i> kckong@kaist.ac.kr
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